

Local AI-Based Adaptive Traffic Signal Control with Emergency Vehicle Priority

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Abstract—Urban traffic congestion is a major challenge caused by increasing vehicle density, fixed-time traffic signals, and unpredictable traffic conditions. Conventional traffic signal systems generally operate using predefined timing plans and may not respond effectively to real-time changes in traffic flow. Recent research has explored artificial intelligence and machine learning techniques for adaptive traffic signal control. However, many intelligent traffic management approaches depend on centralized cloud infrastructure, which can introduce communication delays and dependency on network connectivity.

This paper proposes a conceptual Local AI-Based Adaptive Traffic Signal Control system that performs traffic analysis and signal decision-making near the intersection. The proposed system uses locally collected traffic information to estimate traffic density and dynamically adjust signal timing. An additional emergency vehicle priority mechanism is incorporated to detect emergency situations and provide a temporary priority phase for emergency vehicles. By performing decision-making locally, the proposed approach aims to reduce communication dependency and support faster responses at individual intersections. The paper presents the system architecture, decision-making methodology, mathematical model, algorithm, implementation concept, and evaluation metrics for the proposed system.

Index Terms—Local AI, adaptive traffic signals, intelligent transportation systems, traffic management, emergency vehicle priority, edge computing

I. INTRODUCTION

Rapid urbanization and increasing vehicle ownership have resulted in significant traffic congestion in cities. Traffic intersections are particularly important because inefficient signal timing can lead to unnecessary waiting time, increased fuel consumption, and longer travel durations.

Traditional traffic signals generally use fixed timing schedules. Although these systems are simple to implement, they cannot always respond effectively to rapidly changing traffic conditions. For example, one road may experience heavy traffic while another road has very few vehicles, but a fixed-time controller may continue to allocate similar green-time intervals to both directions.

Artificial intelligence provides an opportunity to develop adaptive traffic signal systems capable of responding to real-time traffic conditions. Reinforcement learning and other machine learning techniques have been investigated for intelligent traffic signal control [1]. Machine vision has also been explored for adaptive traffic light control by analyzing traffic conditions using visual information [2].

Intelligent traffic control can also benefit from local or edge-based processing. Instead of transmitting all traffic information to a remote cloud server, traffic information can be processed closer to the intersection. This can reduce dependency on continuous network connectivity and potentially reduce communication-related delays.

Another important challenge is the movement of emergency vehicles such as ambulances and fire engines. Emergency vehicles may be delayed by congested intersections when normal signal cycles are maintained. Therefore, an adaptive traffic control system should be capable of recognizing emergency situations and temporarily modifying signal operation.

This paper proposes a Local AI-Based Adaptive Traffic Signal Control system with Emergency Vehicle Priority. The proposed system combines local traffic-density analysis with emergency vehicle prioritization to provide adaptive signal decisions at individual intersections.

II. LITERATURE REVIEW

Research on intelligent traffic signal control has expanded significantly with the development of artificial intelligence and machine learning.

Rasheed et al. presented a comprehensive review of deep reinforcement learning techniques for traffic signal control [1]. The study discusses how reinforcement learning can enable traffic signals to adapt their decisions based on traffic conditions. It also highlights challenges associated with intelligent traffic signal control and the use of learning-based approaches.

Gorodokin et al. investigated adaptive traffic light control using machine vision [2]. Their work demonstrates the potential of visual traffic information for optimizing traffic signal operation. Machine vision can provide information about vehicles and traffic conditions without requiring conventional vehicle-counting infrastructure.

Wei et al. proposed IntelliLight, a reinforcement learning approach for intelligent traffic signal control [3]. Their work demonstrates how reinforcement learning can be applied to traffic signal decision-making using traffic-state information.

Although these studies demonstrate the potential of intelligent and adaptive traffic signal control, there remains an opportunity to investigate systems that combine local AI processing with emergency vehicle prioritization. A system

designed to make decisions locally at the intersection can reduce dependence on centralized communication infrastructure while still adapting to changing traffic conditions.

III. PROBLEM STATEMENT

Conventional fixed-time traffic signals are unable to continuously adapt to variations in traffic density. At the same time, centralized intelligent traffic systems may depend on reliable communication networks and cloud infrastructure.

Emergency vehicles can face additional delays when normal signal cycles are maintained during high-priority situations. Therefore, there is a need for an adaptive traffic signal system capable of analyzing local traffic conditions and modifying signal operation when an emergency vehicle requires priority.

The proposed system addresses this problem by using local AI-based traffic analysis together with an emergency vehicle priority mechanism.

IV. RESEARCH GAP

Existing research demonstrates the effectiveness of artificial intelligence, reinforcement learning, and machine vision for adaptive traffic signal control [?], [1], [2].

However, an important research opportunity remains in combining the following characteristics into a single intersection-level system:

- Local traffic analysis and decision-making.
- Adaptive signal timing based on traffic density.
- Reduced dependence on continuous cloud communication.
- Emergency vehicle priority.
- Dynamic coordination of signal phases during emergency situations.

The proposed work focuses on this combination and investigates how local AI can be used to support adaptive traffic signal decisions while giving temporary priority to emergency vehicles.

V. OBJECTIVES

The major objectives of the proposed system are:

- 1) To analyze traffic conditions using locally available traffic information.
- 2) To estimate traffic density at an intersection.
- 3) To dynamically determine appropriate signal timing.
- 4) To identify emergency vehicle situations.
- 5) To provide temporary priority to emergency vehicles.
- 6) To reduce dependence on remote cloud processing.
- 7) To develop a scalable architecture suitable for intelligent transportation systems.

VI. PROPOSED SYSTEM

The proposed system consists of a local sensing and processing unit positioned at or near a traffic intersection. Traffic information can be obtained using cameras, vehicle sensors, or other suitable sensing technologies.

The locally collected information is processed to estimate traffic density. Based on the estimated density, the controller determines the appropriate signal phase and duration.

When an emergency vehicle is detected, the emergency priority module overrides the normal traffic-control decision for a temporary period. The controller can provide a green signal along the required route while maintaining safety constraints for other traffic directions.

The overall system consists of the following major components:

- Traffic sensing module
- Local AI processing module
- Traffic density estimation module
- Emergency vehicle detection module
- Signal decision module
- Traffic light controller

VII. SYSTEM ARCHITECTURE

The conceptual architecture of the proposed system is shown below.

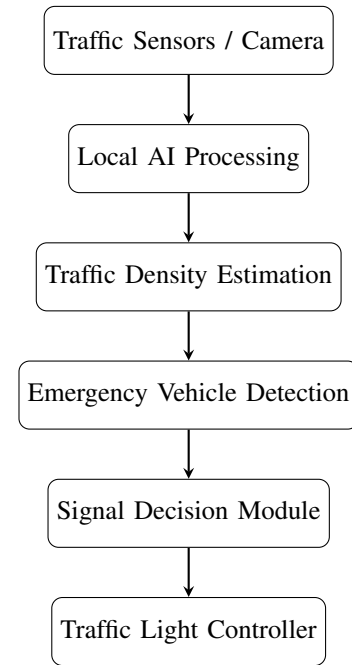


Fig. 1. Conceptual architecture of the proposed local AI traffic signal system.

The sensing module collects information about vehicles and traffic conditions. The local AI processing module analyzes this information without requiring all data to be transferred to a remote server.

The traffic density estimation module determines the relative traffic level for each road. The emergency detection module determines whether an emergency vehicle requires priority.

Finally, the signal decision module selects an appropriate signal phase and sends the decision to the traffic light controller.

VIII. SYSTEM FLOW

The operating procedure of the proposed system is illustrated below.

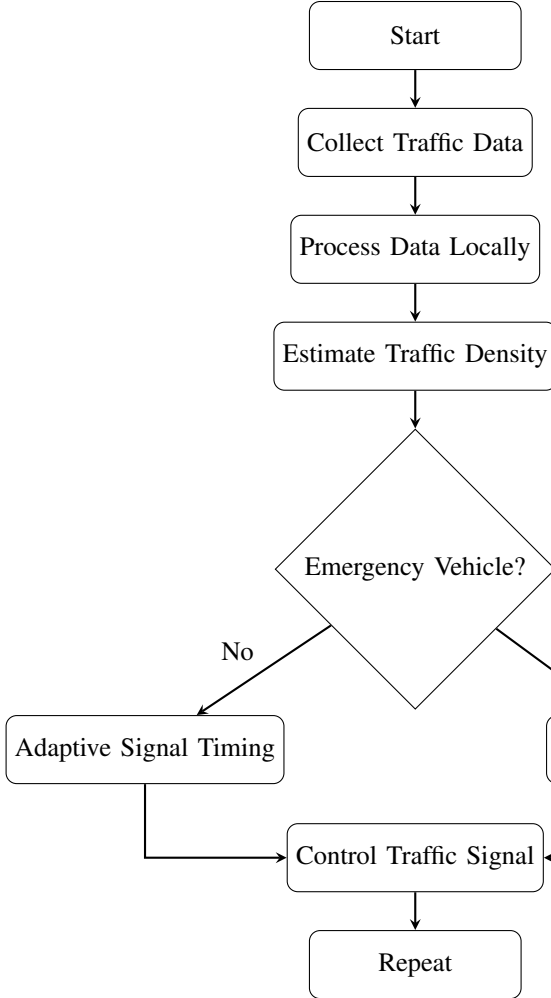


Fig. 2. Flow of the proposed adaptive traffic signal control system.

IX. METHODOLOGY

The proposed methodology consists of five main stages.

A. Traffic Data Collection

Traffic information is collected from cameras or other sensors installed near the intersection. The collected information may include vehicle count, lane occupancy, vehicle movement, and emergency vehicle indicators.

B. Local Data Processing

The collected data is processed locally by an edge computing device or suitable embedded computing platform. Local processing reduces the requirement to transmit raw traffic data continuously to a remote cloud server.

C. Traffic Density Estimation

Traffic density is estimated for each road or lane. A simplified density measure can be represented as:

$$D_i = \frac{N_i}{L_i} \quad (1)$$

where:

- D_i is the estimated traffic density for road i ,
- N_i is the number of detected vehicles,
- L_i represents the effective road or observation length.

The estimated density can be classified into low, medium, or high traffic conditions.

D. Adaptive Signal Decision

The signal controller assigns green time according to traffic conditions. Roads with higher traffic demand may receive longer green intervals, subject to minimum and maximum signal timing constraints.

E. Emergency Vehicle Priority

If an emergency vehicle is detected, the system temporarily changes the normal signal sequence.

An emergency priority variable can be defined as:

$$P_e = \begin{cases} 1, & \text{if an emergency vehicle is detected} \\ 0, & \text{otherwise} \end{cases} \quad (2)$$

When $P_e = 1$, the controller initiates the emergency priority procedure.

Safety constraints should be applied before changing the signal phase to prevent conflicting green signals.

X. ALGORITHM

The basic decision-making procedure is shown in Algorithm 1.

Algorithm 1 Local AI Traffic Signal Control

- 1: Initialize traffic signal controller
 - 2: Collect traffic information
 - 3: Process traffic information locally
 - 4: Estimate traffic density for each road
 - 5: Check for emergency vehicle
 - 6: **if** emergency vehicle detected **then**
 - 7: Activate emergency priority mode
 - 8: Provide safe priority signal phase
 - 9: **else**
 - 10: Calculate adaptive signal timing
 - 11: Select signal phase according to traffic demand
 - 12: **end if**
 - 13: Update traffic signal
 - 14: Repeat the process
-

XI. MATHEMATICAL MODEL

The signal decision can be represented using a traffic demand function:

$$T_i = w_d D_i + w_q Q_i \quad (3)$$

where:

- T_i is the traffic demand score for road i ,
- D_i is the traffic density,
- Q_i represents queue length or waiting vehicles,
- w_d and w_q are weighting factors.

The signal phase with the highest traffic demand can be selected subject to safety and timing constraints.

For emergency situations, the priority decision can be represented as:

$$S_i = \begin{cases} E_i, & P_e = 1 \\ T_i, & P_e = 0 \end{cases} \quad (4)$$

where E_i represents the emergency priority requirement for road i .

XII. IMPLEMENTATION CONCEPT

A prototype implementation can use a camera or traffic sensor for data acquisition and a local computing platform for processing. A lightweight machine learning model can be used for vehicle detection and traffic estimation.

The basic control logic can be represented using Python as follows.

```
def traffic_controller(density, emergency):

    if emergency:
        return "EMERGENCY_PRIORITY"

    if density == "HIGH":
        return "LONG_GREEN"

    elif density == "MEDIUM":
        return "NORMAL_GREEN"

    else:
        return "SHORT_GREEN"

density = "HIGH"
emergency = False

decision = traffic_controller(
    density,
    emergency
)

print(decision)
```

The above code represents the basic decision-making concept. A real implementation would replace the predefined density value with information obtained from sensors or a computer vision model.

TABLE I
CONCEPTUAL COMPARISON OF TRAFFIC SIGNAL APPROACHES

Feature	Fixed-Time Signal	Proposed Local AI System
Traffic adaptation	Limited	Dynamic
Real-time traffic analysis	Limited	Supported
Local processing	Generally not AI-based	Supported
Emergency priority	Usually manual or separate	Integrated
Cloud dependency	Varies	Reduced
Signal timing	Predefined	Adaptive

XIII. CONCEPTUAL COMPARISON

Table I provides a conceptual comparison between conventional fixed-time control and the proposed local AI approach.

The table describes the intended functional differences of the proposed architecture rather than measured experimental performance.

XIV. EXPECTED RESULTS AND DISCUSSION

The proposed system is expected to provide adaptive signal decisions based on the traffic conditions observed at an intersection.

Under low traffic conditions, unnecessary long green intervals can potentially be avoided. Under high traffic conditions, the system can allocate additional green time to the corresponding traffic direction while respecting signal safety constraints.

During emergency situations, the emergency vehicle priority mechanism can temporarily modify the normal signal sequence. This can potentially reduce unnecessary waiting at intersections for emergency vehicles.

Local processing may also reduce dependence on continuous cloud connectivity because traffic information can be analyzed near the intersection.

However, actual improvements in waiting time, travel time, fuel consumption, or emergency response time must be determined through controlled experiments or real-world testing. Such values are not assumed in this conceptual study.

XV. EVALUATION METRICS

The proposed system can be evaluated using the following metrics:

- Average vehicle waiting time.
- Average queue length.
- Traffic throughput.
- Signal decision latency.
- Emergency vehicle waiting time.
- Emergency priority activation accuracy.
- Detection accuracy.
- System resource utilization.

These measurements can be compared with a conventional fixed-time signal controller.

XVI. LIMITATIONS

The proposed system is currently presented as a conceptual architecture and does not report experimentally measured performance.

The accuracy of traffic analysis depends on the quality of the sensing system and the AI model. Poor lighting, occlusion, adverse weather conditions, and dense traffic may affect vehicle detection.

Emergency vehicle detection also requires a reliable identification mechanism. False emergency detection could unnecessarily interrupt normal traffic flow.

Hardware limitations may also restrict the complexity of AI models that can be executed locally. Therefore, model optimization may be required for deployment on resource-constrained edge devices.

XVII. FUTURE SCOPE

Future work can focus on implementing the proposed system using a real traffic intersection or a traffic simulation environment.

A lightweight computer vision model can be deployed on an edge computing device to detect vehicles and estimate traffic density.

Reinforcement learning can also be investigated to allow the signal controller to learn improved timing strategies from traffic conditions.

Another possible extension is multi-intersection coordination. Multiple local AI traffic controllers could exchange limited information with neighboring intersections to coordinate traffic movement while maintaining local decision-making.

Emergency vehicle routing can also be extended to multiple intersections so that a sequence of traffic signals can provide coordinated priority along the emergency vehicle's route.

XVIII. CONCLUSION

This paper proposed a Local AI-Based Adaptive Traffic Signal Control system with Emergency Vehicle Priority. The proposed approach combines local traffic analysis, adaptive signal timing, and emergency vehicle prioritization.

Unlike purely fixed-time traffic signal systems, the proposed approach is designed to respond to changing traffic conditions. Local processing can reduce dependence on continuous communication with remote servers, while the emergency priority mechanism can support faster movement of emergency vehicles through signalized intersections.

The proposed architecture provides a foundation for developing intelligent traffic signal systems using edge computing and artificial intelligence. Future experimental implementation and evaluation are required to quantify improvements in traffic efficiency, emergency response, and system latency.

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